



2400 year old ship found

A ship from the 400 B.C. has been found more than a mile deep in the Black Sea. It's the oldest intact shipwreck known to man. <https://digit.in/nov18-71-1>



AI to create Perfumes

Philyra is an AI solution developed by IBM and Symrise the Master Perfumer. The AI helped create two new fragrances. <https://digit.in/nov18-71-2>

participating countries had internationally set commitment targets, with a legal force. The Paris agreement, also falls under the UNFCCC, but unlike the Kyoto Protocol, has a bottom up structure. The targets are voluntary, and nationally determined. The overall goal is that the rise in global temperatures should be kept well below 2° Celsius, and if possible, below 1.5° C. On 8 October, 2018, the Intergovernmental Panel on Climate Change (IPCC) released a report for policy makers, that shows exactly how the emission reduction goals can be met, and what kind of steps are necessary to do so. The report also highlights the consequences of allowing global temperatures to rise beyond 1.5° C. Hoesung Lee, Chair of the IPCC, said "With more than 6,000 scientific references cited and the dedicated contribution of thousands of expert and government reviewers worldwide, this important report testifies to the breadth and policy relevance of the IPCC."

CLIMATE CRISIS

In the wake of the Paris Agreement, there was a call from scientists for acceptance of a new geological epoch. The world is considered to be in the Holocene epoch since the end of the ice age 10,000 years ago. Normally, geological epochs track to changes that take thousands of years. Since the year 2000, we have pumped more carbon dioxide into the atmosphere than any natural process has managed to do, over the past 800,000 years. With the start of the industrial revolution, humans have drastically changed the atmosphere of the Earth in a very short period of time. Over the past 7000 years, the Earth was seeing a global drop in temperatures at the rate of 0.01° C per century, but since the 1970s, this has turned around sharply, and the global temperatures started rising at the rate of 1.7° C per century. Even drastic natural events have not managed to change the climate as much as humans have. More than geological or biological forces, man shapes

the Earth. Because of this, scientists want to recognise a new geological epoch - the Anthropocene.

The pre-industrial period is the point of comparison for the Paris agreement, and it refers to the time period from 1850 to 1900. Global temperatures have already risen by 1° C since then. We are already facing the consequences, wildfires, melting ice caps, receding coastlines of islands and freak weather conditions. Due to the amount of carbon dioxide we have already put into the atmosphere, and because of ongoing emissions, the global temperatures are rising by about 0.2° C per decade. Here the temperature considered is a global average. In most cases, the temperature of the land has already increased by more than 1.5° C as compared to the pre-industrial time period. Now, even with an increase in temperatures to 1.5° C, humans, and the environment will still have to face plenty of adverse effects of climate change. Bid goodbye to the coral reefs, as most of them will die out. There will be flooding along coastlines, the probability of extreme weather events will increase, unique ecosystems will be threatened, the arctic region will be damaged, and the fishing industry will suffer. The sea levels will continue to rise beyond 2100, but the rate at which this happens can be limited, if the temperature rise is kept in check. We will still see animal extinction and biodiversity loss, but the projected numbers for these is significantly lower at 1.5° C as compared to 2° C. Terrestrial, coastal and freshwater ecosystems are the ones most likely to be adversely affected.

India will face the most adverse effects of climate change, even if the necessary measures are put in place to curtail the emissions. India is going to see intense category 4 and category 5 tropical cyclones. The coastlines will rise. We are expected to see issues with food security till the year 2300. Remember the deadly heatwave of 2015? We can expect more such heatwaves, even if the temperature rise is kept below 1.5° C. The number of draughts are

projected to increase. In fact, there are already some drylands in India where the temperatures have risen by over 1.5° C, and the people living here are moving away. We will see more such climate change driven emigration.

GLOBAL WARNING

At this point, the scientists have already accepted that humanity is going to face widespread, adverse effects of climate change. The oceans are going to become increasingly acidic, consequently affecting marine life. The health, livelihood and food supply of humans are all going to suffer with a global increase in temperature of less than 1.5° C. Go beyond 2° C, and all of that will just be much, much worse.

Simply put, the whole world is not doing enough to keep the rise in temperatures in check. If all the countries around the world do not come together to reduce emissions, then the writing is on the wall. We are facing a man made disaster of colossal proportions. Humans need to press the reset button, and rethink everything. We need to change the way we build cities, use land, generate electricity, the kind of transport we use, and how we manufacture things. Not only do the changes need to be drastic, they need to be fast. By 2030, we need to reduce the amount of carbon dioxide emissions by 45 percent, as compared to 2010. By 2050, the emissions should reduce to net zero. If we start pumping in any more carbon dioxide into the atmosphere, then we have to start actively taking it out of the air. We still do not know if this is even possible. The situation is so bad, that scientists are actively thinking of far fetched solutions such as painting the mountain tops white, and sending giant mirrors into space.

Priyadarshi Shukla, Co-Chair of IPCC Working Group III said, "This report gives policymakers and practitioners the information they need to make decisions that tackle climate change while considering local context and people's needs. The next few years are probably the most important in our history." 